ARCHANA VERMA

Urbana, IL | (224) 279-5171 | archana3@illinois.edu | www.linkedin.com/in/archana-verma

EDUCATION

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Ph.D. Candidate | Chemical Engineering | May 2026 GPA: 3.95

M.S. | Chemical Engineering | May 2024

Fellowships: *NSF Graduate Research Fellow, Illinois Distinguished Fellow, Samuel W. Parr Fellow*

STANFORD UNIVERSITY

B.S. | Chemical Engineering | June 2021 GPA: 3.86

RESEARCH EXPERIENCE

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Graduate Research Assistant | Prof. Nicholas E. Jackson | Theoretical Materials Chemistry | Sept. 2021 - Present

- Engineered a custom C/C++ molecular simulation framework from the ground up to integrate quantum chemistry and materials modeling for doped organic semiconductors; created and led cross-functional subgroup for doping team
- Developed machine learning workflow collaborating with experimentalists to extract design rules for chiral polymers
- Challenged conventional assumptions of how disorder affects charge transfer using quantum mechanical analysis (DFT)

STANFORD UNIVERSITY

Undergraduate Research Assistant | Prof. Gerald G. Fuller | Complex Fluids | Sept. 2019 - June 2021

- Consulted with industry partner (Johnson & Johnson) to enhance contact lens product performance through data-driven recommendations and rheometric image analysis (MATLAB) of tear film stability

Undergraduate Research Assistant | Prof. Michael D. Fayer | Physical Chemistry | Sept. 2017 - June 2019

- Led the development and characterization of novel material frameworks to investigate fundamental molecular mechanisms of proton transport, applying state-of-the-art 2D spectroscopy to solve complex chemical challenges

INTERNSHIPS

BASF

PhD Leadership Development Program Intern | Process Operations Analyst | June 2025 - Aug. 2025

- Spearheaded laboratory analysis and strategic implementation for a new process step to improve lubricant additive conductivity, scaling operations from lab to pilot plant on an international team

CORTEVA AGRISCIENCE

R&D Intern | Formulations Chemistry | June 2020 - Aug. 2020

- Initiated cross-functional collaboration between chemistry and informatics teams to engineer an automated simulation workflow, delivering an accessible software solution (Python) for component material screening

GEORGIA INSTITUTE OF TECHNOLOGY

Research Intern | Prof. Jesse G. McDaniel | Computational Chemistry | June 2018 - Aug. 2019

- Independently analyzed molecular simulation data (Python) to identify thermodynamic driver of ionic liquid miscibility

REGENERON PHARMACEUTICALS

R&D Intern | Analytical Chemistry | June 2017 - Aug. 2017

- Implemented new protocol to improve contaminant detection in pre-clinical drug testing of monoclonal antibodies

NEW YORK UNIVERSITY

Research Intern | Prof. Daniel B. Turner | Physical & Inorganic Chemistry | June 2015 - Aug. 2016

- Engineered high-precision optical experiments to redefine theoretical models of electronic relaxation

OUTREACH & LEADERSHIP

AAAS Catalyzing Advocacy in Science and Engineering (CASE) Workshop | Attendee | 2026

Graduate Society of Women Engineers | weSTEM Director, Speaker Coordinator, Social Media Chair | 2023 - Present

- Led executive-level management of 11-member team for weSTEM conference (130 attendees and 5 corporate partners)
- Directed program development for 23 diverse speakers, overseeing lifecycle of initial selection to scheduling operations
- Executed digital branding strategy yielding exponential growth in reach (3,400%) and engagement (2,200%)
- Earned Grassroots Initiatives to Address Needs Together Student Leadership Award for impactful community change

Crisis Text Line | Volunteer Crisis Counselor | 2023 - Present

- Applied de-escalation and collaborative problem-solving to manage high-pressure situations

Try Engineering Together | eMentor | 2023 - Present

- Championed talent development by designing career roadmaps, fostering an inclusive pipeline for junior scientists

Zindaa UIUC | Choreographer and Instructor | 2022 - Present

- Directed instruction for an 18-member team, managing creative and operational workflows for group performances

SciSquare Scholars | Diversity in Science and Engineering Program Mentor | 2020 - 2021

- Coached junior researchers through project design and academic planning, fostering professional growth

American Institute for Chemical Engineers | Peer Mentor | 2020 - 2021

- Provided career consulting and resource management tools for undergraduate students within chemical engineering

Stanford University Physics Society | Advertising & Diversity Chair | 2018 - 2020

- Launched university-wide educational series and directed multi-platform digital communication strategies

PUBLICATIONS

** denotes equal contribution*

[8] **Verma, A.,** Wang, C.-I., *et. al.* (2025). Conformation Mediated Doping in P3HT:F4TCNQ Dimers from Density Functional Theory. *Journal of Physical Chemistry A. 29*(44). 10108-10116. https://doi.org/10.1021/acs.jpca.5c05040

[7] Raghuraman, V., **Verma, A.**, & Jackson, N.E. (2025) All-Atom Reactive Monte Carlo Molecular Dynamics for Molecular Doping in Organic Semiconductors. *Journal of Chemical Theory and Computation. 21*(20). 10527-10536. https://doi.org/10.1021/acs.jctc.5c01206

[6] Xu, Z.*, Wang, Z.*, Zhang, R.*, **Verma, A.***, *et. al.* (2025). Ubiquitous Chiral Symmetry Breaking of Conjugated Polymers via Liquid-Liquid Phase Separation. *Journal of the American Chemical Society. 147*(37), 33530-33544. https://doi.org/10.1021/jacs.5c07995

[5] **Verma, A.,** & Jackson, N. E. (2024). Assessing Molecular Doping Efficiency in Organic Semiconductors with Reactive Monte Carlo. *Journal of Chemical Physics. 160*(10), 104106. https://doi.org/10.1063/5.0197816

[4] Suja, V. C.*, **Verma, A.***, *et. al.* (2022). Dewetting Characteristics of Contact Lenses Coated with Wetting Agents. *Journal of Colloid and Interface Science.* (614), 24-32. https://doi.org/10.1016/j.jcis.2022.01.075

[3] **Verma, A.**, Stoppelman, J.P., & McDaniel, J.G. (2020). Tuning Water Networks via Ionic Liquid/Water Mixtures. *International Journal of Molecular Sciences. 21*(2), 403-427. https://doi.org/10.3390/ijms21020403

[2] McDaniel, J. G., & **Verma, A.** (2019). On the Miscibility and Immiscibility of Ionic Liquids and Water. *Journal of Physical Chemistry B, 123*(25), 5343–5356. https://doi.org/10.1021/acs.jpcb.9b02187

[1] Carbery, W. P., **Verma, A.**, & Turner, D. B. (2017). Spin-Orbit Coupling Drives Femtosecond Nonadiabatic Dynamics in a Transition Metal Compound. *Journal of Physical Chemistry Letters, 8*(6), 1315-1322. https://doi.org/10.1021/acs.jpclett.7b00130

PRESENTATIONS & AWARDS

University Archives Women in Science Lecture | Oral (Invited) | 2026

American Institute for Chemical Engineers (AIChE) Annual Conference | Poster | 2020, 2025
- 2025: Women in Chemical Engineering Community Travel Award winner

Gender Equity and Inclusion in Chemical Sciences Conference | Panelist | 2024
- Gender Equity and Inclusion in Chemical Sciences Inclusive Leadership Award Winner

American Physical Society (APS) March Meeting | Oral | 2024

Midwest Thermodynamics and Statistical Mechanics Conference | Poster | 2023 | Oral | 2024

UIUC Chemical Engineering Graduate Student Symposium | Poster | 2023 | Oral | 2024
- 2024: First Place Oral Presentation

American Chemical Society (ACS) Central Regional Meeting | Oral (Invited) | 2023

Regeneron Science Talent Search | Oral | 2017
- Fifth Place Overall Nationally

University of Toronto International Research Invitational | Oral (Invited) | 2017

Intel International Science and Engineering Fair | Poster | 2016, 2017
- 2017: Second Place in Chemistry, ACS Third Place Special Award
- 2016: First Place in Chemistry

New York State Science and Engineering Fair | Poster | 2016, 2017
- 2017: First Place in Chemistry, Millennial Materials Special Award
- 2016: First Place in Chemistry

Long Island Science and Engineering Fair | Poster | 2016, 2017
- 2017: First Place in Chemistry & Materials Science, ASM Materials Science Special Award
- 2016: First Place in Chemistry, Yale Science and Engineering Association Special Award

National Junior Science and Humanities Symposium | Poster | 2016
- Second Place in Chemistry Poster Division

TEACHING EXPERIENCE

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Graduate TA | Data Science for Chemistry and Engineering | Fall 2023
- Independently led weekly office hours, graded coding assignments, and was responsible for course communications

Graduate TA (Technical Writing Team) | Process Design | Fall 2022
- Co-led weekly discussion sections and office hours focused on enhancing technical communication for capstone design

STANFORD UNIVERSITY

Undergraduate TA | Foundations of Physical Chemistry | Spring 2021
- Created course homeworks and exams, held weekly office hours, managed communications, grading, and Q&A forum

Undergraduate TA | Physical Chemistry I: Thermodynamics | Spring 2020
- Co-led two weekly computational labs, held weekly office hours, managed communications, grading, and Q&A forum

SKILLS & COURSEWORK

Programming & Software: Python, C/C++, MATLAB, Mathematica, Java, NWChem, LAMMPS, Avogadro
Scientific: Molecular Simulation, Data Science, Physical Chemistry, Polymer Physics, Optical Spectroscopy
Selected Coursework: Chemical Processes in Energy Systems, Understanding Energy, Bioenergy and Biofuels, Transportation Equity